

STAT 341: Tutorial Test 1 - Sample 2

First name

Last name

Signature

ID Number

Email

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

Please read: this practice test is designed to give you a clear sense of the sort of things that can come up in exams. Please note that this is, of course, only a subset of the material that could be covered. I would **strongly** encourage you to attempt this test under exam conditions - do not be tempted to look up something in your notes until afterwards! This is one reason why it is common for students to find real exams much harder than practice ones, even though the material is of an equivalent difficulty!

1. (5 marks) Consider the following measure of spread:

$$\alpha_N(y_1, \dots, y_N) = \left(\max_i (y_i) - \min_i (y_i) \right) = y_{(N)} - y_{(1)}$$

(a) (3 marks) Derive this statistic's sensitivity curve:

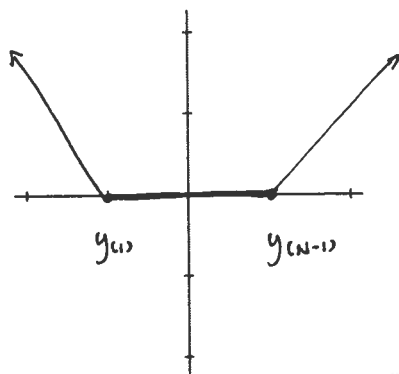
$$\text{Let } \alpha_{N-1}(y_1, \dots, y_{N-1}) = y_{(N-1)} - y_{(1)}.$$

$$\alpha_N(y_1, \dots, y_{N-1}, y) = \begin{cases} y_{(N-1)} - y & \text{if } y < y_{(1)} \\ y_{(N-1)} - y_{(1)} & \text{if } y \in [y_{(1)}, y_{(N-1)}] \\ y - y_{(1)} & \text{if } y > y_{(N-1)} \end{cases}$$

$$\text{So, } SC(y) = N(\alpha_N(y_1, \dots, y_{N-1}, y) - \alpha_{N-1}(y_1, \dots, y_{N-1}))$$

$$SC(y) = \begin{cases} N(y_{(1)} - y) & \text{if } y < y_{(1)} \\ 0 & \text{if } y \in [y_{(1)}, y_{(N-1)}] \\ N(y - y_{(N-1)}) & \text{if } y > y_{(N-1)} \end{cases}$$

(b) (1 mark) Give a sketch of the sensitivity curve: (c) (1 mark) From this curve, what do you conclude about this statistic's resistance to outliers in y ?



It's not resistant against outliers, since the sensitivity curve is unbounded.

2. Answer the following short questions:

(a) [2 marks] Provide two differences between BGD: Batch Gradient Descent and SGD: Stochastic Gradient Descent algorithms.

1. BGD computes gradient on all N units in population before updating θ .
SGD updates θ after computing gradient on each unit.

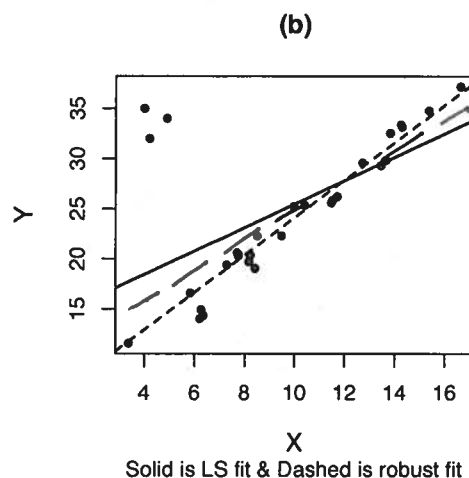
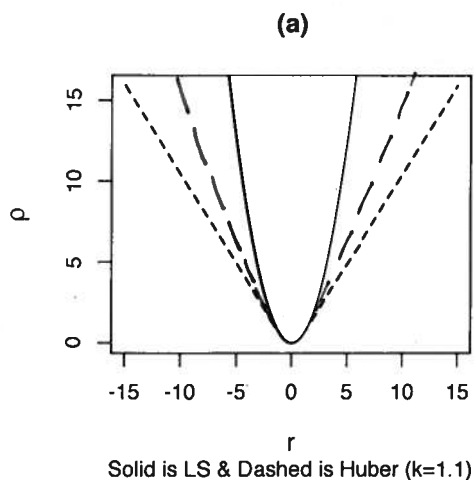
2. Sample/Solution path of SGD is usually noisier than that of BGD.

(b) [2 marks] Provide one example each of implicitly and explicitly defined attributes.

Explicit: mean = $\frac{1}{N} \sum_{u \in \mathcal{P}} y_u$

Implicit: least-square parameter: $\hat{\theta} = \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} (y_u - \theta)^2$

3. [2 marks] The plots below show the ρ functions along with the fitted lines for least squares and for robust regression using Huber's ρ with $k = 1.1$. The explanations are at the bottom of each panel. In panel (a), sketch the Huber's ρ function for $k = 3$, and in panel (b), draw the fitted robust regression line using Huber's ρ function with $k = 3$.



4. In a population $\mathcal{P} = \{(x_1, y_1), \dots, (x_N, y_N)\}$, consider the problem of calculating β for the model $y_u = \beta x_u + r_u$. This model is called *regression through origin* (the intercept is set to 0). An example of data where this model would be applicable is the relationship between brain weight and body weight in mammals, where body weight = 0 implies brain weight = 0. We would like to calculate the parameter β of this model using

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}} \sum_{u=1}^N \rho(r_u)$$

where $\rho(r) = r^4$. We are interested in using Newton's algorithm to compute the value of $\hat{\beta}$.

- (a) [3 marks] Derive the ψ function for Newton's method, and write the equation (in terms of β and data) to which Newton's algorithm should find a solution.

$$\rho(\beta) = \sum_{u=1}^N (y_u - \beta x_u)^4 \Rightarrow \psi(\beta) = \frac{\partial}{\partial \beta} \rho(\beta) = -4 \sum_{u=1}^N x_u (y_u - \beta x_u)^3$$

$$N-R \text{ attempts to solve } \psi(\beta) = 0 \text{ or } \sum_{u=1}^N x_u (y_u - \beta x_u)^3 = 0.$$

- (b) [3 marks] In bullet points, provide the steps of Newton's algorithm to calculate β in this particular problem. Any function/derivative should be calculated explicitly as the ρ function has been provided in the question.

$$\text{Firstly, obtain } \psi'(\beta) = \frac{\partial}{\partial \beta} \psi(\beta).$$

$$\psi'(\beta) = 12 \sum_{u=1}^N x_u^2 (y_u - \beta x_u)^2$$

Now the algorithm:

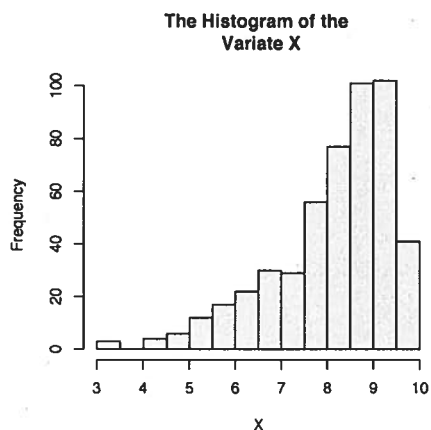
Initialize: $i=0$ & initial estimate $\hat{\beta}_0$

$$\text{LOOP: 1. Update parameter } \hat{\beta}_{i+1} = \hat{\beta}_i - \frac{\left(-4 \sum_{u=1}^N x_u (y_u - \hat{\beta}_i x_u)^3\right)}{\left(12 \sum_{u=1}^N x_u^2 (y_u - \hat{\beta}_i x_u)^2\right)}$$

2. Test for convergence: If converged: STOP LOOP
else: $i=i+1$, REPEAT LOOP

$$\text{Output: } \hat{\beta} = \hat{\beta}_i$$

5. [2 marks] The histogram of a variate x in a population is plotted below. Which of the following transformations may make the distribution more symmetric? Circle all that apply.



(i) $\frac{1}{x}$

(ii) $\sqrt{x}$

(iii) x^2

(iv) x^4

(v) $\log(x)$

6. [4 marks] Determine whether the following statements are true or false.

(a) There exist more than 150 possible samples, each of size 7, from a population of size 10.

i. True

ii. False

(b) The gradient descent algorithm is a root finding algorithm.

i. True

ii. False

(c) Consider all possible samples of size n (where $n < N$) from the population $\mathcal{P} = \{y_1, \dots, y_N\}$. The sample error for the range attribute ($\max_i y_i - \min_i y_i$) is often negative.

i. True

ii. False

(d) If $\alpha_1(\mathcal{P})$ and $\alpha_2(\mathcal{P})$ are two attributes, both location-scale equivariant, then their average, i.e.

$$\alpha(\mathcal{P}) = \frac{\alpha_1(\mathcal{P}) + \alpha_2(\mathcal{P})}{2}$$

is also a location-scale equivariant attribute.

i. True

ii. False

